

VOIDSCOPE · DESIGN DECISIONS REGISTER

VoidScope design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/voidscope		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Register opened at TRL 3; every decision made under Amish's 2026-10-03 pre-approval

VoidScope design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

Safety: Several decisions below set safety stops: the drip ceiling, the bite valve, the blower's shut-off pressure and the battery chemistry. They were taken on the conservative side (VDS-DDR-001, Table 2). VoidScope is rescue equipment that is not certified and is not a medical device.

■ Open decisions

None. All decisions were made under Amish's 2026-10-03 pre-approval.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The camera head is 29 mm or less across and about 55 mm long, IP68 at 1 m for 30 min, AHD 1080p with about a 90° view and 12 dimmable LEDs	It sets the nose bore, R3 and R4	VDS-DDR-001 D4; VDS-CAL-001 C
2	The blower's shut-off (maximum static) pressure is 1.0 kPa or less, and its curve gives at least 20 L/min through 0.38 kPa	It is the air pressure safety stop (R6)	VDS-DDR-001 D8, S3; VDS-CAL-001 E
3	The flow meter's pressure drop is 0.2 kPa or less at 20 L/min and it fits 100 mm tall	A larger drop lowers the full-speed flow; the lid must close	VDS-CAL-001 E; VDS-DDR-002 P11
4	The snap button suits 1.47 mm and 2.11 mm walls, its button stands about 1.5 mm proud, and it holds a 300 N pull	It locks every joint	VDS-DDR-002 P2; VDS-CAL-001 B
5	The two tube sizes telescope with about 0.11 mm a side as bought	The couplers must slide in by hand	VDS-DDR-002 P1
6	The capillary bore is 0.5 mm and the measured flow at 1 m and 20 °C is about 1.5 mL/min	It sets the drip ceiling (R5)	VDS-DDR-001 S1; VDS-CAL-001 D
7	The bite valve is about 11 mm across, fits the 4 mm tube and stays shut under 20 kPa of water head	It is the water safety stop	VDS-DDR-001 S2; VDS-CAL-001 D7
8	The speaker is 20 mm across and 4 mm thick, IP67	It must sit flush in its pocket	VDS-DDR-002 P8
9	The hybrid cable is 6 mm or less across, and the case, monitor (28 mm deep) and battery (151 x 65 x 94 mm) match the sizes modelled	They set the grommets, the plate layout and the lid stack	VDS-DDR-002 P10, P11

■ Value engineering

Value-engineering target: USD 2,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 1,578 (USD 422 under the target). Main cost drivers and savings worth trying:

- The largest lines are the lathe time allowance (USD 180), the monitor with recorder and bracket (USD 175), the cleaning kit and four spare water sets (USD 118), the rod sections (USD 128 for four), the rod carry case (USD 115), the battery packs (USD 110 for two), the surface unit case (USD 110) and the camera head (USD 95).
- Savings worth trying: a builder with a lathe saves USD 180; a soft carry bag in place of the hard rod case saves about USD 80; a single battery pack saves USD 55 but leaves no spare for a full shift at -10 °C.
- Making the design constructable added the couplers, buttons and pins, the guide tube, the equipment plate, the brackets and the lathe allowance, about USD 290 in all.

■ Decisions made

Date	Decision	Decided by	Record
2026-10-03	TRL 2 review items D1 to D13: lines inside with the bore as air duct; 38.1 mm aluminium sections with snap buttons; no articulation; bought IP68 camera head; monitor in the surface unit lid; tip speaker as speaker and microphone; gravity drip with capillary ceiling and bite valve; blower chosen by shut-off pressure; two LiFePO4 packs; independent of the crawler block; requirements restated; budget, pitch and problem kept	Amish: "I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."	VDS-DDR-001
2026-10-03	Conservative safety choices S1 to S6, each with the evidence that would relax it	Amish, same pre-approval	VDS-DDR-001, Table 2
2026-10-03	First co-design candidate to approach: the disaster management training school of the Armed Police Force, Nepal; second: Turkey's AFAD training centres (not contacted, not partners)	Amish, same pre-approval	VDS-DDR-001 D11
2026-10-03	Design for construction, changes P1 to P14	Amish, same pre-approval	VDS-DDR-002
2026-10-03	R11 (two 45° bends) recorded as not met in the first version; a flexible lead section is a later version, not TRL 3 work	Amish, same pre-approval	VDS-REQ-001 v0.3; VDS-DDR-001 D12
2026-10-03	Appearance model departures from <code>model.py</code> (rounded edges, depth bands, labels, cables drawn to the surface unit, rubble context) accepted for renders only	Amish, same pre-approval	<code>docs/</code> <code>REVIEW.md</code> , TRL 3